SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

TOTAL:98/100

COURSE CODE: CSA4303

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.

2. Identify the **appropriate HTTP method for form submission** and explain why. 3. Choose the **proper semantic element for navigation** and justify its usage.

4. Identify the **correct XML syntax** and explain the error in other options. 5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I **M.Nirosha(192311411)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M.Nirosha

RUBRICS FOR CALCULATION:

Criteria Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case 25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts 25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis 19	In-depth analysis with strong reasoning	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

	and insights	
Solution Design 19	Innovative, feasible solutions with strong justification Logical solutions with adequate justification Basic solutions with weak justification	Incorrect or no solution provided
Clarity 10	Clear, well structured, and easy to understand Organized and understandable presentation Limited clarity and structure	Poor clarity; difficult to understand

TOTAL:98/100

Web Development Standards Implementation

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

1. Select the correct HTML tag for video embedding and justify your answer. Option B — <video src="video.mp4" controls></video>

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Video Embedding</title>
</head>
<body>
  <h2>Promotional Video</h2>
  <video src="video.mp4" controls width="400">
```

```
    Your browser does not support video playback.
  </video>
</body>
</html>
```



Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

2. Identify the appropriate HTTP method for form submission and explain why.
Option B — POST

Code:

```
<!DOCTYPE html>

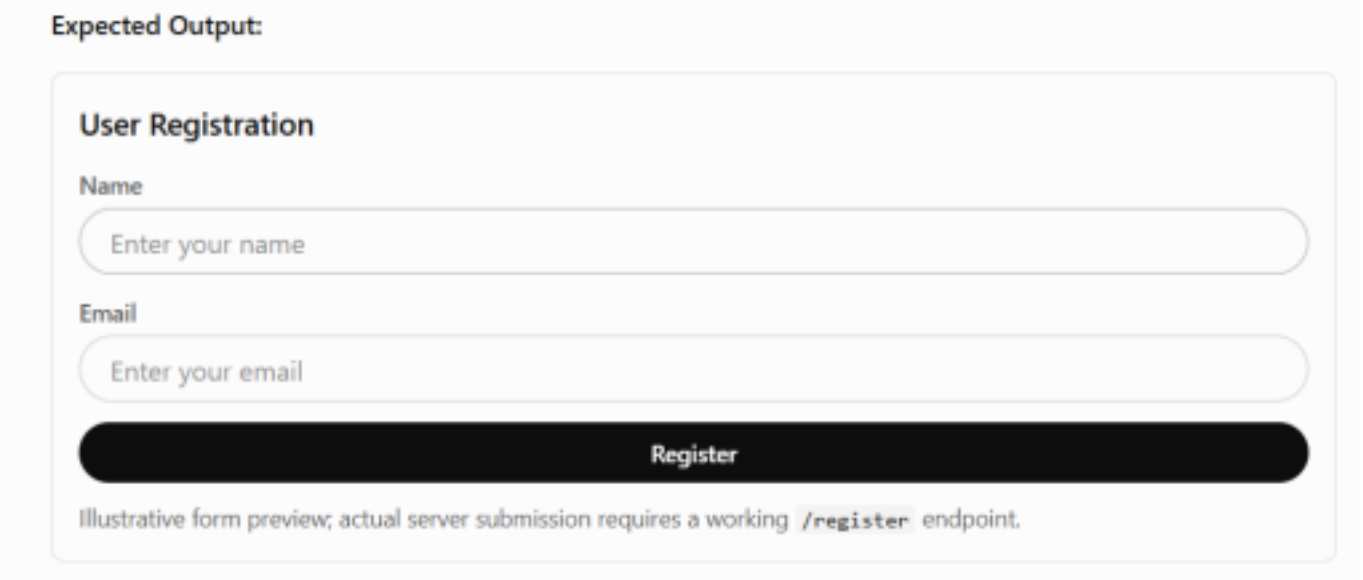
<html lang="en">
<head>

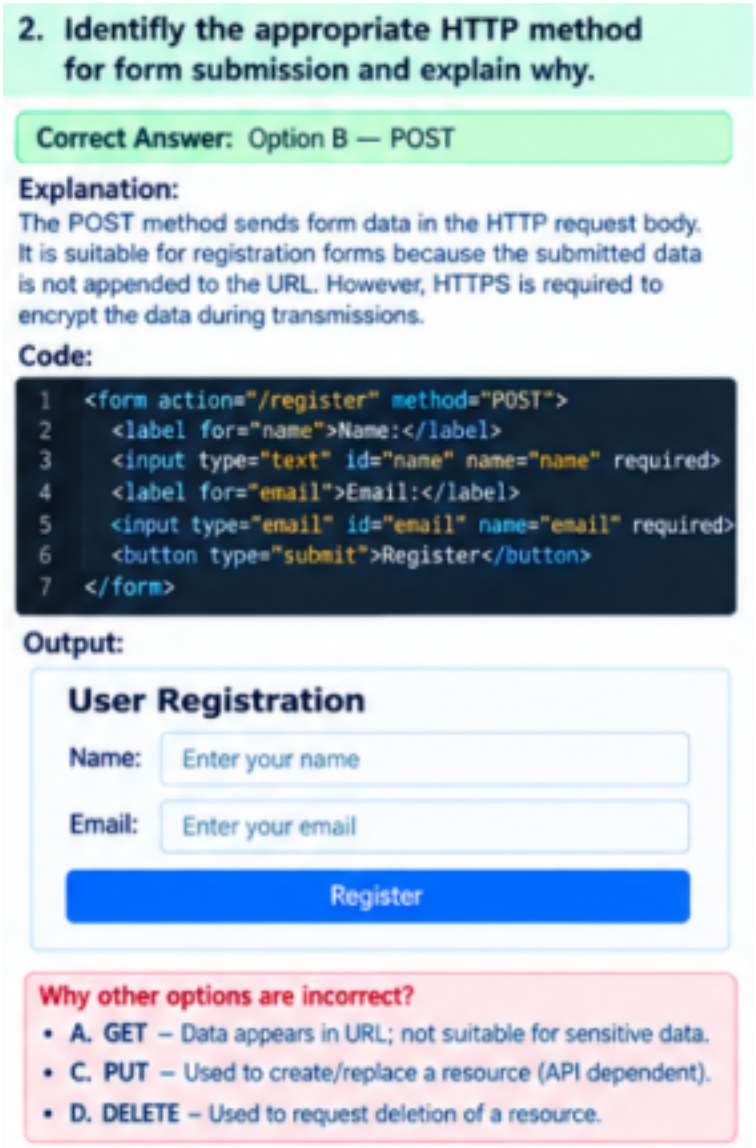
  <meta charset="UTF-8">

  <title>Registration Form</title>
```

```
</head>
<body>
  <h2>User Registration</h2>
  <form action="/register" method="POST">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required>
    <br><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>
    <br><br>
    <button type="submit">Register</button>
  </form>
</body>
</html>
```

Expected Output:





Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

3. Choose the proper semantic element for navigation and justify its usage. Option B — <nav>

Code:

<!DOCTYPE html>

<html lang="en">

<head>

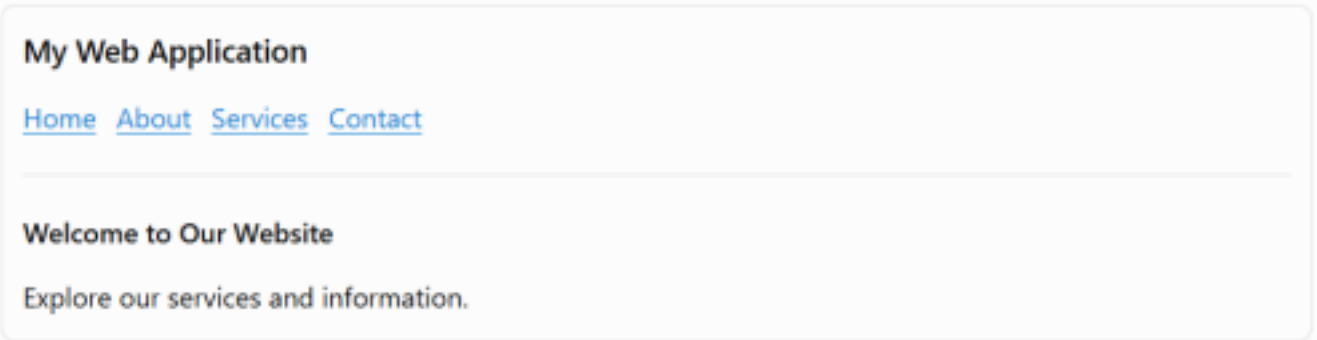
<meta charset="UTF-8">

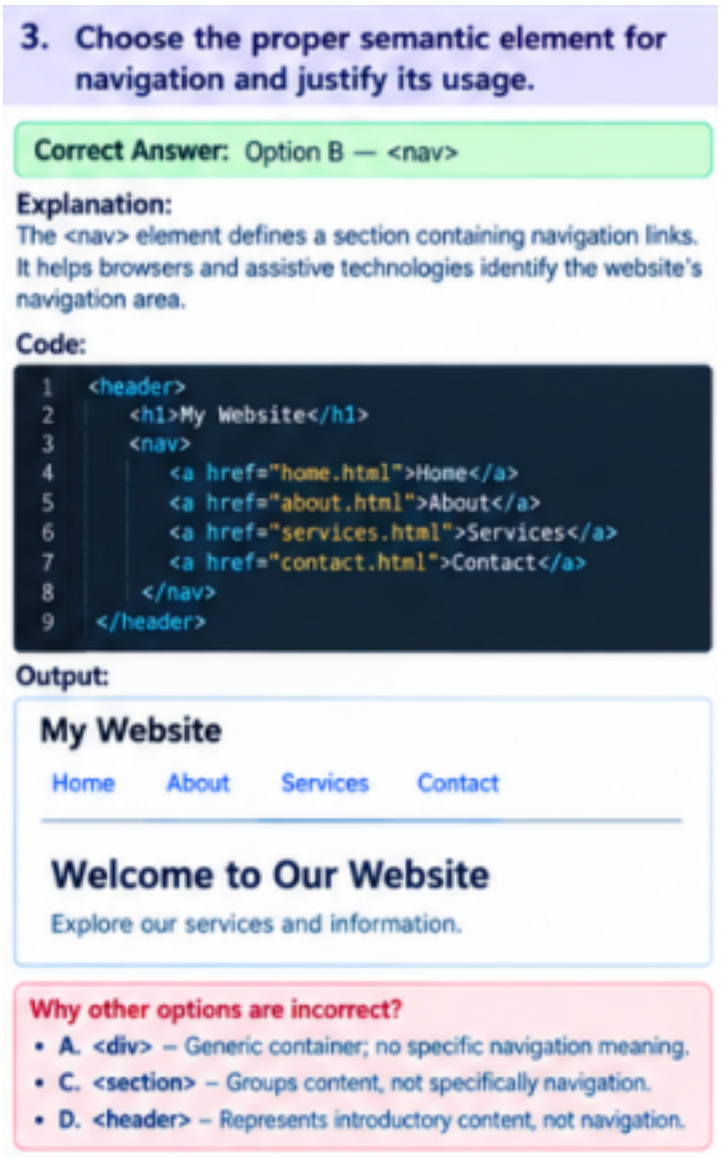
<title>Navigation Menu</title>

</head>

```
<body>
    <header>
        <h1>My Web Application</h1>
        <nav aria-label="Main navigation">
            <a href="#home">Home</a>
            <a href="#about">About</a>
            <a href="#services">Services</a>
            <a href="#contact">Contact</a>
        </nav>
    </header>
    <main>
        <h2 id="home">Welcome to Our Website</h2>
        <p>Explore our services and information.</p>
    </main>
</body>
</html>
```

Expected Output:





Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

4. Identify the correct XML syntax and explain the error in other options. Option B:

<note><to>User</to><from>Admin</from></note>

Code:

```
<?xml version="1.0" encoding="UTF-8"?>
<note>

   <to>User</to>

   <from>Admin</from>

</note>
```



Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

5. Choose the correct protocol for web communication and justify your choice. Option B — HTTP

Code:

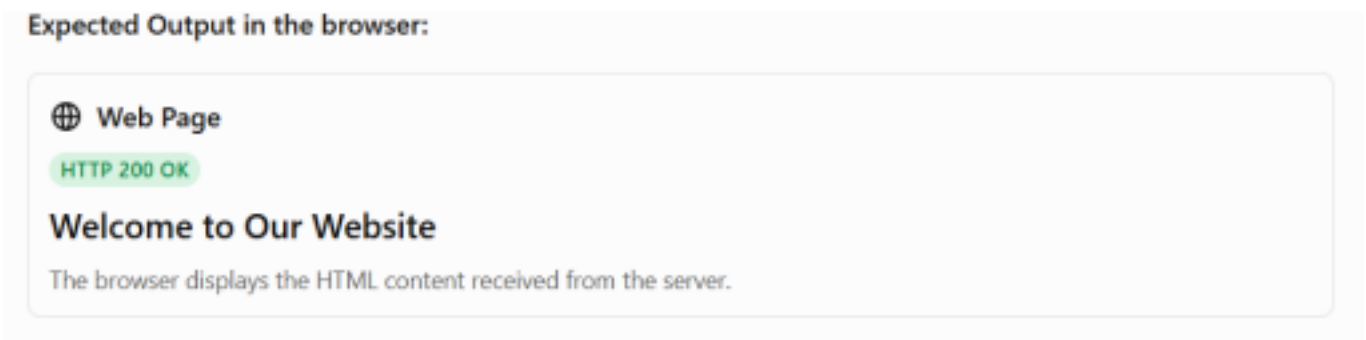
GET /index.html HTTP/1.1

Host: example.com

Accept: text/html

HTTP/1.1 200 OK
Content-Type: text/html

<h1>Welcome to Our Website</h1>



5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: Option B — HTTP

Explanation:
HTTP (Hypertext Transfer Protocol) is used for communication between a web client and server. The client sends an HTTP request and the server returns an HTTP response. For secure communication, HTTPS (HTTP over TLS) is used.

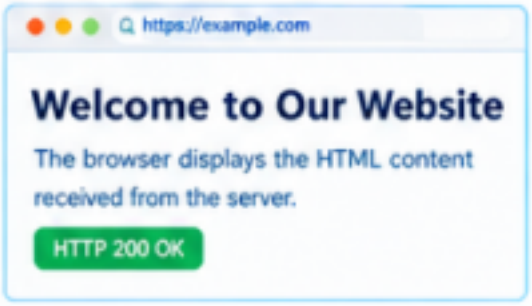
HTTP Request and Response Example:

```
GET /index.html HTTP/1.1
Host: example.com
Accept: text/html

HTTP/1.1 200 OK
Content-Type: text/html

<html>
    <body>
      <h1>Welcome to Our Website</h1>
    </body>
</html>
```

Expected Output in Browser:



Why other options are incorrect:

- **A. FTP** – Used for file transfer.
- **C. SMTP** – Used for sending email.
- **D. SNMP** – Used for monitoring network devices.